



PRODUCT NAME

Hard-Water Dishwashing Detergent Tablet Formula

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Short Description

High-quality detergent formulations - powder - powder with low viscosity, ph 7, medium quality cost level, special requirements: fabric softening, stain removal for professional use

Technical Specifications

pH Level:	7.0
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100 kg
Processing Time:	2 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Ingredient	INCI Name	%	Function
Water	Aqua	85.0%	Base solvent
Glycerin	Glycerin	10.0%	Humectant
Preservative	Phenoxyethanol	3.0%	Preservation
Fragrance	Parfum	2.0%	Scent

Manufacturing Process

Phase 1: Main Phase

1. Combine all ingredients
2. Mix thoroughly
3. Package

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, and lab coat.
PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.
Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.
Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.